

full-fledged automation solution, many other parameters are needed to be looked at. These include kind of applications, number of orders needed to be served, kind of SKUs that are involved and so on. Hence, more than cost, it is important to evaluate whether the solution a business is planning to invest in is what it really needs in terms of results.

Having said that, a typical estimation of the upfront investment for such automation solutions can start from US\$ 300,000 to US\$ 400,000, depending on the scale of use. Businesses should look at solution features like sustainability, scalability, complexity in maintenance, risk of obsolescence, after-sales support and the like.

Q. How quickly can they recover these investments?

A. Typically, as in the case of Butler, the ROI varies from three to six years, depending on geographies, use cases, scale of applications and type of ROI involved, such as faster order-to-dispatch, accuracy and customer experience.

Q. Why has the adoption of these technologies been slower in India as compared to the rest of the world?

A. In India, the general technology adoption is a little slow, which is a challenge. Moreover, investing in a platform like Butler makes sense only when there is a lot of complexity in the business. Complexities may come for different reasons. It could be the same warehouse serving through several channels, that is, the omnichannel environment, or serving orders with very high SKUs, or because of huge warehouses having very high throughput load.

Indian warehouses are maturing, but they are yet to face all these complexities in full form. For instance, many players have already started exploring the omnichannel model. They have recognised that it will become a dominating trend in future retail.

Moreover, when GST was not in place, larger players had broken down their operations into multiple warehouses because of state taxes. With GST coming in, businesses have started merging their smaller warehouses into bigger ones. We have data that shows a trend of about 50 per cent of businesses on the verge of merging their operations into bigger warehouses. This will increase throughput volumes and SKUs, and

drive the demand for automated technologies.

However, this will not happen overnight. The trend and demand will set in with time.

Q. Can SMBs leverage on these technologies, too?

A. Every automation requires a certain business scale. SMBs have to make sure they have sufficient scale of requirement. Having said that, we are seeing more and more players considering these solutions. Players whom we connected with a couple of years ago, but who were not ready to invest in these technologies back then, are reaching out to us now, showing interest in automation.

Q. How do you ensure the safety of human peers who work with robots?

A. Safety is of paramount importance. In Butler there are several levels of safety. For instance, if someone comes in front of Butler while it is moving, it senses the person and stops immediately. It also has an earthquake feature, using which it automatically stops operating when it senses an earthquake—to ensure the shelves do not topple.

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GREEN TECH:

“People In The Country Do Not Have In-Depth Knowledge About The **VARIOUS TYPES** Of **AIR POLLUTANTS**”

➤ Air quality has become a matter of concern in urban Indian cities, like Kolkata and New Delhi, that suffer from hazardously high air quality index (AQI) consistently. To address this problem, administrations and consumers alike need to stay updated about the air conditions in real time. Rohit Bansal, founder, Purelogic Labs India, talks to Paromik Chakraborty of *Electronics For You* about how national air quality monitors, like their AQI India platform, can help us take the first step towards better health

Q. What are the benefits of deploying large-scale air quality monitoring platforms like AQI India in the country?

A. Platforms like AQI India not only help everyone know the real-time AQI

values but also let them be cautious about pollution trends of different locations. Along with detailed information on pollution parameters, including measurement of PM 2.5 and PM 10, these provide health advisories based

on the AQI of a location. The platforms are accessible by anyone.

Q. What are the technical elements that enable this system?

A. AQI monitors are placed at more